

Relational Algebra & SQL Examples

Exercise 4.3.

Suppliers(*sid*: integer, *sname*: string, *address*: string)

Parts(*pid*: integer, *pname*: string, *color*: string)

Catalog(*sid*: integer, *pid*: integer, *cost*: real)

The key fields are underlined, and the domain of each field is listed after the field name. Therefore *sid* is the key for Suppliers, *pid* is the key for Parts, and *sid* and *pid* together form the key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers. Write the following queries in relational algebra and SQL:

1. Find the *names* of suppliers who supply some red part.

Suppliers(sid: integer, sname: string, address: string)

Parts(pid: integer, pname: string, color: string)

Catalog(sid: integer, pid: integer, cost: real)

$$\pi_{sname}(\pi_{sid}((\pi_{pid} \sigma_{color='red'} Parts) \bowtie Catalog) \bowtie Suppliers)$$

```
SELECT S.sname
FROM   Suppliers S, Parts P, Catalog C
WHERE  P.color='red' AND C.pid=P.pid AND C.sid=S.sid
```

2. Find the *sids* of suppliers who supply some red or green part.

Suppliers(sid: integer, sname: string, address: string)

Parts(pid: integer, pname: string, color: string)

Catalog(sid: integer, pid: integer, cost: real)

$$\pi_{sid}(\pi_{pid}(\sigma_{color='red' \vee color='green'} Parts) \bowtie catalog)$$

```
SELECT C.sid
FROM   Catalog C, Parts P
WHERE  (P.color = 'red' OR P.color = 'green')
       AND P.pid = C.pid
```

3. Find the *sids* of suppliers who supply some red part or are at 221 Packer Street.

Suppliers(sid: integer, sname: string, address: string)

Parts(pid: integer, pname: string, color: string)

Catalog(sid: integer, pid: integer, cost: real)

$\rho(R1, \pi_{sid}((\pi_{pid} \sigma_{color='red'} Parts) \bowtie Catalog))$	SELECT S.sid
$\rho(R2, \pi_{sid} \sigma_{address='221PackerStreet'} Suppliers)$	FROM Suppliers S
$R1 \cup R2$	WHERE S.address = '221 Packer street'
	OR S.sid IN (SELECT C.sid
	FROM Parts P, Catalog C
	WHERE P.color='red' AND P.pid = C.pid)

4. Find the *sids* of suppliers who supply some red part and some green part.

Suppliers(sid: integer, sname: string, address: string)

Parts(pid: integer, pname: string, color: string)

Catalog(sid: integer, pid: integer, cost: real)

$\rho(R1, \pi_{sid}((\pi_{pid} \sigma_{color='red'} Parts) \bowtie Catalog))$

$\rho(R2, \pi_{sid}((\pi_{pid} \sigma_{color='green'} Parts) \bowtie Catalog))$

$R1 \cap R2$

```
SELECT C.sid
FROM   Parts P, Catalog C
WHERE  P.color = 'red' AND P.pid = C.pid
      AND EXISTS ( SELECT P2.pid
                    FROM   Parts P2, Catalog C2
                    WHERE  P2.color = 'green' AND C2.sid = C.sid
                        AND P2.pid = C2.pid )
```


6. Find the *sids* of suppliers who supply every red part.

Suppliers(*sid*: integer, *sname*: string, *address*: string)

Parts(*pid*: integer, *pname*: string, *color*: string)

Catalog(*sid*: integer, *pid*: integer, *cost*: real)

```
SELECT C.sid
FROM   Catalog C
WHERE  NOT EXISTS (SELECT P.pid
                   FROM   Parts P
                   WHERE  P.color = 'red'
                   AND (NOT EXISTS (SELECT C1.sid
                                   FROM   Catalog C1
                                   WHERE  C1.sid = C.sid AND
                                         C1.pid = P.pid)))
```

$(\pi_{sid, pid} Catalog) / (\pi_{pid} \sigma_{color='red'} Parts)$

8. Find the *sids* of suppliers who supply every red part or supply every green part.

Suppliers(sid: integer, sname: string, address: string)

Parts(pid: integer, pname: string, color: string)

Catalog(sid: integer, pid: integer, cost: real)

$\rho(R1, ((\pi_{sid, pid} Catalog) / (\pi_{pid} \sigma_{color='red'} Parts)))$
 $\rho(R2, ((\pi_{sid, pid} Catalog) / (\pi_{pid} \sigma_{color='green'} Parts)))$
 $R1 \cup R2$

```
SELECT C.sid
FROM   Catalog C
WHERE  (NOT EXISTS (SELECT P.pid
                    FROM   Parts P
                    WHERE  P.color = 'red' AND
                    (NOT EXISTS (SELECT C1.sid
                                FROM   Catalog C1
                                WHERE  C1.sid = C.sid AND
                                      C1.pid = P.pid))))
OR ( NOT EXISTS (SELECT P1.pid
                FROM   Parts P1
                WHERE  P1.color = 'green' AND
                (NOT EXISTS (SELECT C2.sid
                            FROM   Catalog C2
                            WHERE  C2.sid = C.sid AND
                                  C2.pid = P1.pid))))
```

9. Find pairs of *sids* such that the supplier with the first *sid* charges more for some part than the supplier with the second *sid*.

Suppliers(*sid*: integer, *sname*: string, *address*: string)

Parts(*pid*: integer, *pname*: string, *color*: string)

Catalog(*sid*: integer, *pid*: integer, *cost*: real)

$\rho(R1, \text{Catalog})$

$\rho(R2, \text{Catalog})$

$\pi_{R1.sid, R2.sid}(\sigma_{R1.pid=R2.pid \wedge R1.sid \neq R2.sid \wedge R1.cost > R2.cost}(R1 \times R2))$

```
SELECT C1.sid, C2.sid
FROM   Catalog C1, Catalog C2
WHERE  C1.pid = C2.pid AND C1.sid  $\neq$  C2.sid
      AND C1.cost > C2.cost
```

10. Find the *pids* of parts supplied by at least two different suppliers.

Suppliers(sid: integer, sname: string, address: string)

Parts(pid: integer, pname: string, color: string)

Catalog(sid: integer, pid: integer, cost: real)

$\rho(R1, Catalog)$

$\rho(R2, Catalog)$

$\pi_{R1.pid} \sigma_{R1.pid=R2.pid \wedge R1.sid \neq R2.sid} (R1 \times R2)$

```
SELECT C.pid
FROM   Catalog C
WHERE  EXISTS (SELECT C1.sid
                FROM   Catalog C1
                WHERE  C1.pid = C.pid AND C1.sid  $\neq$  C.sid )
```

11. Find the *pids* of the most expensive parts supplied by suppliers named Yosemite Sham.

Suppliers(sid: integer, sname: string, address: string)

Parts(pid: integer, pname: string, color: string)

Catalog(sid: integer, pid: integer, cost: real)

$\rho(R1, \pi_{sid} \sigma_{sname='YosemiteSham'} Suppliers)$

$\rho(R2, R1 \bowtie Catalog)$

$\rho(R3, R2)$

$\rho(R4(1 \rightarrow sid, 2 \rightarrow pid, 3 \rightarrow cost), \sigma_{R3.cost < R2.cost}(R3 \times R2))$

$\pi_{pid}(R2 - \pi_{sid,pid,cost} R4)$

SELECT C.pid

FROM Catalog C, Suppliers S

WHERE S.sname = 'Yosemite Sham' AND C.sid = S.sid

AND C.cost $\geq$ ALL (Select C2.cost

FROM Catalog C2, Suppliers S2

WHERE S2.sname = 'Yosemite Sham'

AND C2.sid = S2.sid)